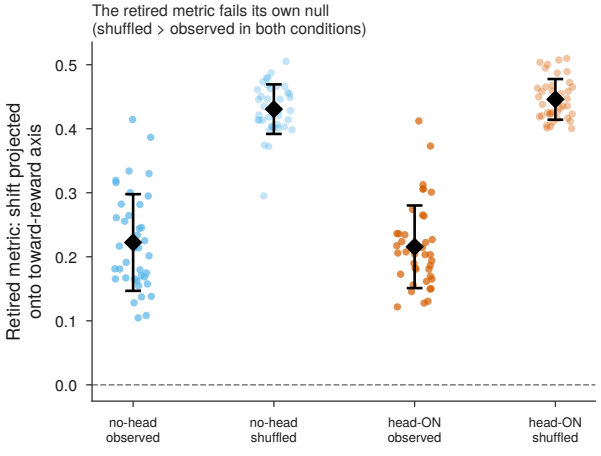
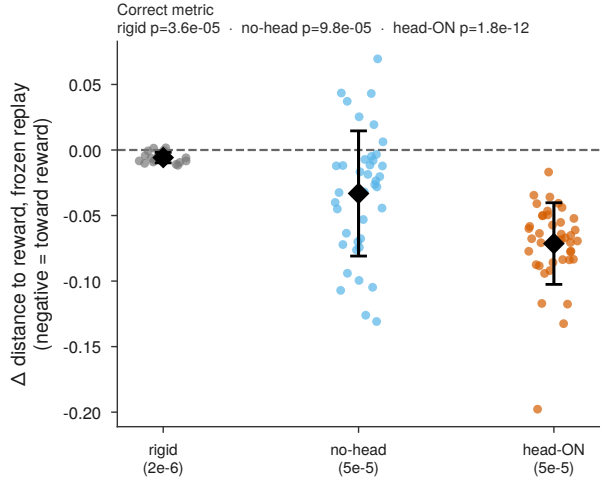
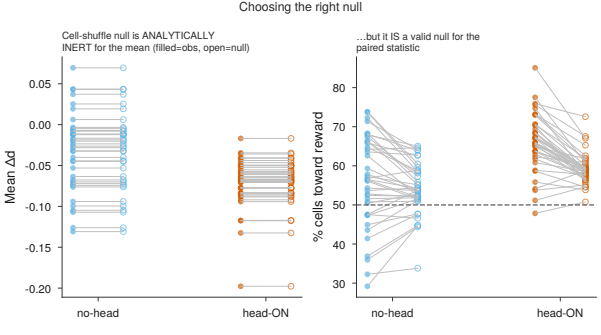
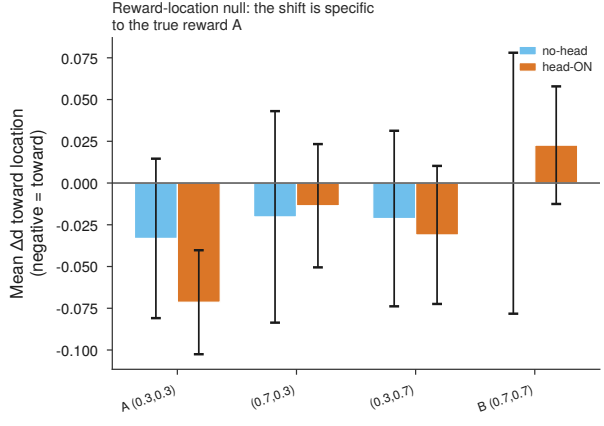

The retired metric fails its own null
(shuffled > observed in both conditions)
Retired metric: shift projected
onto toward-reward axis
0.5
0.4
0.3
0.2
0.1
0.0
no-head
observed
no-head
shuffled
head-ON
observed
head-ON
shuffled


Correct metric
rigid p=3.6e-05 · no-head p=9.8e-05 · head-ON p=1.8e-12
Δ distance to reward, frozen replay
(negative = toward reward)
0.05
0.00
-0.05
-0.10
-0.15
-0.20
rigid
(2e-6)
no-head
(5e-5)
head-ON
(5e-5)


Choosing the right null
Cell-shuffle null is ANALYTICALLY
INERT for the mean (filled=obs, open=null)
Mean Δd
0.05
0.00
-0.05
-0.10
-0.15
-0.20
no-head
head-ON
...but it IS a valid null for the
paired statistic
% cells toward reward
80
70
60
50
40
30
no-head
head-ON


Reward-location null: the shift is specific
to the true reward A
Mean Δd toward location
(negative = toward)
no-head
head-ON
0.075
0.050
0.025
0.000
-0.025
-0.050
-0.075
-0.100
A (0.3,0.3)
(0.7,0.3)
(0.3,0.7)
B (0.7,0.7)